Wissal Eddalia

AI & Data Science Engineer

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PROFESSIONAL SUMMARY

Final-year AI & Data Science engineering student at ENSIA with hands-on experience in machine learning, deep learning, NLP, computer vision, reinforcement learning, and Geo-AI. Built production-style full-stack AI systems (FastAPI, React, PostgreSQL/PostGIS), led development teams, and won national hackathons. Open to research collaborations, internships, and full-time roles.

EDUCATION

National School of Artificial Intelligence (ENSIA) 2022 – 2027

Engineering Degree in AI & Data Science, Sidi Abdellah, Algiers, Algeria

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Speech Processing, Computer Vision, Time Series Analysis, Robotics, Reinforcement Learning, Wireless Communication, Full-Stack Development, Databases, Statistical Computing, Data Mining, Algorithms, Optimization, Distributed Systems

PROFESSIONAL EXPERIENCE

Final-Year Project – LLM-Based API Integration Assistant, ENSIA – Co-supervised by University of Nantes & Salvia Oct 2026 – Jun 2027

Ensia , Algiers, Algeria

- Conducting final-year research project (FYP/2026/244) on translating natural-language integration requests into executable API workflows.
- Working with Large Language Models (LLMs), OpenAPI/Swagger specifications, workflow generation, data mapping, and webhook configuration.
- Developing a framework for workflow validation, error handling, and human-in-the-loop correction to improve reliability and user control.
- Supervised by Prof. Fouad Dahak (ENSIA), Prof. Dalila Tamzalit (University of Nantes), and Jean-Philippe Gouigoux (Salvia).

Research Intern & AI Developer, Algerian Space Agency (ASAL) Sep 2025 – Mar 2026

Algiers, Algeria

- Completed the Advanced Geo-AI & Space Technology Program and built ACIETS, Algeria's first sovereign satellite-based greenhouse gas (GHG) monitoring platform.
- Served as Development Team Leader: designed FastAPI microservices, PostgreSQL/PostGIS schema, and React/Mapbox frontend.
- Fine-tuned STARCOP U-Net for methane plume detection; integrated ANN and Random Forest forecasting models.
- Continued as active researcher in ASAL's incubator, contributing to ongoing Geo-AI projects.

Software Development Intern, Algerie Poste Dec 2025 – Mar 2026

Algiers, Algeria

- Built an internal dynamic audit questionnaire platform with AI-powered insights and workflow automation.
- Architected questionnaire logic, an automated scoring engine, and AI-generated audit recommendations.

AI & Data Science Intern, Djezzy 2025

Algiers, Algeria

- Built a Geographic Segmentation System for telecom antenna placement using clustering, NLP, and classification, with a React/GeoJSON interface.
- Performed end-to-end Time Series Forecasting of sales, network traffic, and revenue using ARIMA and Prophet models.

PROJECTS

Research Production Analysis System 2025

Technologies: NLP, Transformers, SPECTER2

- LLM-powered scholarly impact platform; fine-tuned SPECTER2 on 1.4M papers.
- Computed a Relative Impact Index (RII) normalizing citations within domain-specific peer groups to address h-index field bias.

ACIETS – Algerian Carbon Intelligence & Emissions Tracking 2025
Technologies: FastAPI, React, PostGIS, PyTorch

- Full-stack national-scale GHG monitoring platform: 11-table database, 5 microservices, 25+ React/Mapbox components, STARCOP fine-tuning, anomaly detection, and multi-horizon forecasting.

Deep Reinforcement Learning – Adaptive Traffic Signal Control 2026
Technologies: Python, PyTorch, SUMO, Stable-Baselines3

- Developed a deep RL traffic signal controller in SUMO, modeling a four-way intersection as an MDP with a 38-dimensional state and adaptive phase selection.
- Trained and compared DQN, PPO, and A2C; PPO achieved an 80.4% reduction in average waiting time and 59.6% lower queue length while maintaining baseline throughput.

Sanad: Evidence-Grounded Claim Verification 2026
Technologies: LLMs, NLP, RAG, FastAPI, React

- Built an LLM-powered fact-checking system that retrieves and ranks evidence sources to classify online claims with evidence-backed verdicts and explanations.
- Developed a Manifest V3 browser extension and FastAPI backend, integrating structured verification outputs, graceful error handling, and claim-specific media-literacy tips and quizzes.

AI Search – Supply Chain and Logistics Optimization 2024
Technologies: Python, BFS, A, Folium*

- Optimized transport routes across Algerian wilayas using BFS, UCS, A*, Hill Climbing, and CSP with interactive Folium map visualizations.

ML & Data Analysis – Classification, Regression & Time Series 2025
Technologies: Python, R, scikit-learn, SARIMA

- Implemented Decision Trees, Random Forests, KNN, SVM, and Neural Networks for car price prediction and image classification.
- Conducted time series analysis of U.S. Natural Gas Production (1973–2024) using R and SARIMA; delivered energy policy insights.
- Developed a sleep stage classification system from EEG signals; engineered 57 domain-specific features and trained SVM models.

Web & Mobile – Mzia.dari and Loue 2024
Technologies: React, Node.js, Supabase, Flutter

- Mzia.dari: job matching platform built with React, Node.js, and Supabase following Scrum methodology.
- Loue: car rental mobile application developed in Flutter.

Computer Vision – Satellite Object Detection (Hackathon) 2024
Technologies: YOLOv8, Remote Sensing

- Developed 3 computer vision projects (stadium, greenhouse, and palm tree detection) using YOLOv8 on satellite imagery; won 3rd place at LASAL/SecAI Hackathon.

Cybersecurity Educational Tools 2025
Technologies: Python, Desktop App

- Built an Expense Tracker with Ransomware Simulation for awareness training, and a Password Security Checker for evaluating credential strength.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, TypeScript, Dart, R, SQL
Web Technologies: HTML, CSS, React, FastAPI, Flutter, Node.js
AI & Data Science: Machine Learning, Deep Learning, Natural Language Processing, Speech Processing, Computer Vision, Time Series Analysis, Reinforcement Learning, Data Mining, Statistical Computing
Data Management: Database Design, Database Management, Data Modeling, PostgreSQL, PostGIS
Engineering & Computing: Algorithms, Optimization, Distributed Systems, Robotics, Wireless Communication
Frameworks & Tools: PyTorch, scikit-learn, Stable-Baselines3, Git, Docker, QGIS
Productivity Tools: Microsoft Word, Excel, PowerPoint

AWARDS

- **Djezzy Ideathon Winner** – AI-driven solution Dec 2025
- **3rd Place, LASAL/SecAI Hackathon** – Computer vision with satellite imagery 2024
- **Baccalaureate with Honors** – 17.67/20 2022

LEADERSHIP & ACTIVITIES

- **Communication Department Manager**, SecAI ENSIA Community 2025 – 2026
- **Development Team Leader**, Algerian Space Agency (ASAL) – Led the development team of ACIETS, a satellite-based greenhouse gas monitoring platform. 2025 – 2026
- **Team Lead**, 3 academic projects – coordination, delivery, and timeline management
- **Hackathon Participant** – AI Fest, HackTrain, Agri Challenge

LANGUAGES

Arabic (Native), English (Fluent), French (Fluent)

INTERESTS

AI for Software Engineering, Applied ML Systems, Reinforcement Learning, NLP & LLMs, Computer Vision, Time Series Analysis, Remote Sensing & Geo-AI, AIOps & Observability